



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RAMCHANDER ENTERPRISES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 100.40 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (11.591059, 78.611191)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 1, Attur - Perambalur Rd, Attur, encompasses approximately 100.4 kilometers along a mix of national and state highways. The predominant path includes NH79 and SH30, which offer direct connectivity.

Typical Weather Conditions and Hazards

This region experiences a hot semi-arid climate with high temperatures, especially between March and June. Monsoon season lasts from June to September, bringing moderate to heavy rainfall that could lead to flooding or water-logging in low-lying areas. Between December and February, cooler temperatures prevail with minimal rainfall. Monitor weather forecasts for rain and prepare for potential reduced visibility and slippery road conditions.

Traffic Patterns

Traffic congestion is generally experienced near urban centers such as Salem and diversion points near Kallakurichi. Peak hours are typically from 8:00 AM to 10:00 AM and 6:00 PM to 8:00 PM. NH79 often experiences higher traffic volume, especially near toll booths and junctions, causing delays.

Road Quality and Infrastructure

The road quality is largely good along NH79, with periodic maintenance ensuring a smooth ride. However, SH30 might have some sections with potholes and uneven surfaces. Watch for construction zones and diversions that can cause sudden changes in traffic flow.

Alternative Routes for Emergencies

In emergencies, consider:

- Taking NH44 if you need to bypass a blocked NH79 section.
- Utilizing local roads parallel to the primary route, although these may have lower quality and higher traffic.

Local Regulations on Hazardous Materials

Tamil Nadu enforces strict regulations on transporting hazardous materials, including specific permits and adherence to safety standards. Restrictions are in place for travel through certain sensitive areas, especially near residential zones.

Historical Incidents

There have been occasional accidents involving heavy vehicles on NH79, mainly due to over-speeding and driver fatigue. Incidents involving hazardous materials remain rare but underscore the importance of compliance with safety regulations.

Environmental Considerations

Be cautious in areas near water bodies and agricultural lands to prevent contamination from spills. The region is also home to some ecologically sensitive zones that restrict hazardous material transit.

Communication Coverage

Cellular reception is generally reliable along NH79, with potential dead zones in more remote sections of SH30. Carry emergency communication devices for uninterrupted connectivity.

Emergency Response Times

Response times in urban areas or near towns are faster, averaging 15-30 minutes. In remote stretches, expect delays with times potentially extending to an hour. Note the locations of hospitals and emergency services along the route for quick access.

Overall Summary of Risk Assessment

The route is mostly straightforward with well-maintained highways, although weather and traffic can introduce risks. Ensure vehicles are properly maintained and drivers are aware of peak congestion times, local regulations, and alternative routes. Adequate preparation and awareness can significantly minimize the risks associated with transporting hazardous materials on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.64140, 78.11801	30 KM/Hr
13	Turn	Medium	11.64179, 78.11822	30 KM/Hr
14	Turn	Medium	11.64218, 78.11881	30 KM/Hr
15	Turn	Medium	11.64192, 78.12133	30 KM/Hr
16	Turn	Medium	11.64156, 78.12162	30 KM/Hr
17	Turn	Medium	11.64116, 78.12168	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	Medium	11.63934, 78.12082	30 KM/Hr
19	Turn	Medium	11.65154, 78.35007	30 KM/Hr
20	Blind Spot	Blind Spot	11.59767, 78.61845	10 KM/Hr
21	Turn	High	11.59846, 78.61161	15 KM/Hr
22	Turn	High	11.59652, 78.61065	15 KM/Hr
23	Turn	High	11.59626, 78.61111	15 KM/Hr
24	Turn	Medium	11.59503, 78.61074	30 KM/Hr
25	Turn	High	11.59285, 78.61014	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
24	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
25	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
26	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
27	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
28	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
29	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
30	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
31	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
32	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
33	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
39	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
41	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium
42	hospital	Amsh Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium
43	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium
44	hospital	Government Hospital, Kothampadi	11.62641, 78.542199	30 km/h	Medium
46	hospital	Lakshitha Hospital	11.5952715, 78.6094785	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
34	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
35	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium
36	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
37	school	Sri Vidhya Mandir Senior Secondary School	11.668421, 78.2509573	30 km/h	Medium
38	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
40	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium
45	school	Vani Vikas Matric Hr.sec school	11.6212713, 78.5462658	30 km/h	Medium

Route Photos of Risky Spots